

ST BENEDICT'S COLLEGE



SUBJECT	APM Paper 1	DATE	12 November 2016
GRADE	11	MARKS	180
EXAMINER	Mrs Serafino	MODERATOR	Mrs Povall & Mrs Sillman & Mr Benecke
NAME		DURATION	2.5 hours
CLASS			

QUESTION	ASS STANDARD DESCRIPTION	TOTAL	ACTUAL
1 – 4	Algebra	77	
5	Trigonometry	17	
6 – 7	Calculus and Differentiation	50	
8	Integration	15	
9 – 10	Application of integration	22	
	TOTAL	180	

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

- 1) This question paper consists of 7 pages.
- 2) Read the questions carefully.
- 3) Answer all the questions.
- 4) Number your answers as the questions are numbered.
- 5) All the necessary working details must be clearly shown.
- 6) Approved non-programmable calculators may be used except where otherwise stated.
- 7) It is in your own interest to write legibly and to present your work neatly.
- 8) Where necessary, round off to two decimal places.

QUESTION 1**29 MARKS**Solve for $x \in \mathcal{R}$; without the use of a calculator:

a) $x^3 = 2x(x + 4)$ 6

b) $|2x^2 - x| - 1 + \frac{15}{|2x^2 - x| + 2} = 5$ 8

c) $\frac{2x-3}{4} \geq \frac{4}{2x+3}$ 8

d) $\frac{-3}{x^2} + 6 + \frac{17}{x} \leq 0$ 7

QUESTION 2**18 MARKS**

a) Simplify: $\sqrt{i^4}$ 2

b) Find the real values of a and b such that $(3 + 2i)(a + 3i) = bi$ 7

c) If $x = 2i$ and $x = 3 - i$ are solutions of the quartic equation (a 4th degree equation) $f(x) = x^4 + ax^3 - bx^2 + cx + d$, find the values of a, b, c and d . 9

QUESTION 3**11 MARKS**

Three functions whose symbolic expressions look rather similar are listed below:

$$f(x) = \frac{1}{x-3}$$

$$g(x) = \frac{1}{|x-3|}$$

$$h(x) = \frac{1}{\sqrt{x-3}}$$

a) For which values of x are each one of the functions above undefined? 3

b) Make rough sketches of each of the functions on separate sets of axes. 6

c) Is there an axes of symmetry in any of the graphs? If so, express the axis of symmetry as an equation. 2

QUESTION 4

19 MARKS

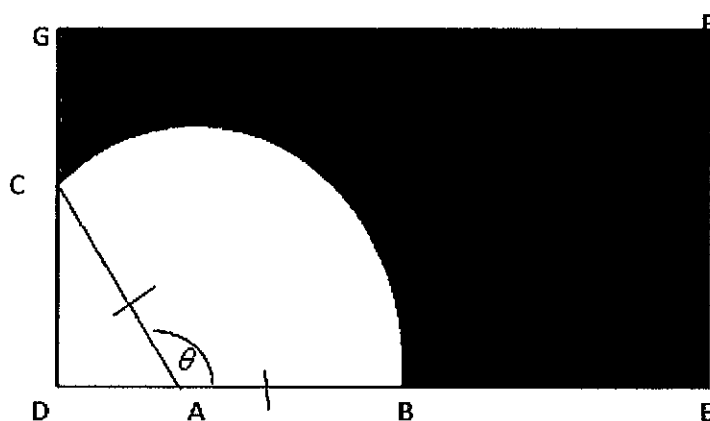
- a) On the same set of axes sketch: $f(x) = 2|x + 2| - 3$ and $g(x) = \left| \frac{3}{x} \right|$ 6
- b) Hence find the exact solution to: $2|x + 2| - 3 \geq \left| \frac{3}{x} \right|$ 8
- c) It is given that $f(x) = x^2 - 6x + 5$ and $g(x) = |x|$.
Sketch: $y = g(f(x))$ 5

QUESTION 5

17 MARKS

- a) Prove the identity: $(\sec \alpha - \cos \alpha)^2 = \frac{1}{\cot^2 \alpha} - \sin^2 \alpha$. 8

b)



A goat is tied to a pole at A with a rope of 2 meters in length in a rectangular enclosure DEFG with dimensions 5m x 3m.

If $AB = AC$ and $AD = 1$ m; determine

- 1) Show that the size of $\theta = \frac{2\pi}{3}$. 2
- 2) The shaded area which represents the area where the goat cannot reach. 7
(Do not simplify your answer)

QUESTION 6

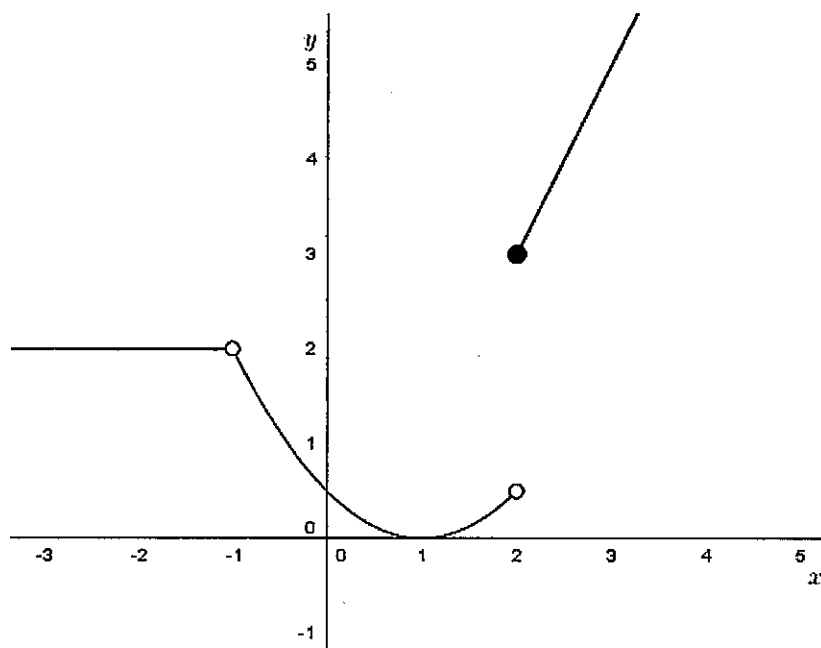
23 MARKS

a) State whether each of the following statements is TRUE or False:

- 1) If a function is differentiable at a point, then the limit of the function must exist at that point.
- 2) If a function is continuous at a point, then it must also be differentiable at that point.
- 3) At a local maximum, the gradient of the graph is decreasing.

6

b)



If it is given that $f(x) = \begin{cases} 2 & \text{if } x < -1 \\ \frac{1}{2}(x-1)^2 & \text{if } -1 < x < 2 \\ 2x-1 & \text{if } x \geq 2 \end{cases}$

- 1) Give values of x where the function is discontinuous and state the type of discontinuity. (Reasons not required)

4

- 2) i) Find $f(-2)$

1

- ii) Find $f(0)$

1

- iii) Find $f(2)$

1

- 3) Is the curve differentiable at $x = 2$? Give a reason for your answer

2

- 4) Prove that the curve is differentiable at $x = 1$

8

QUESTION 7**27 MARKS**

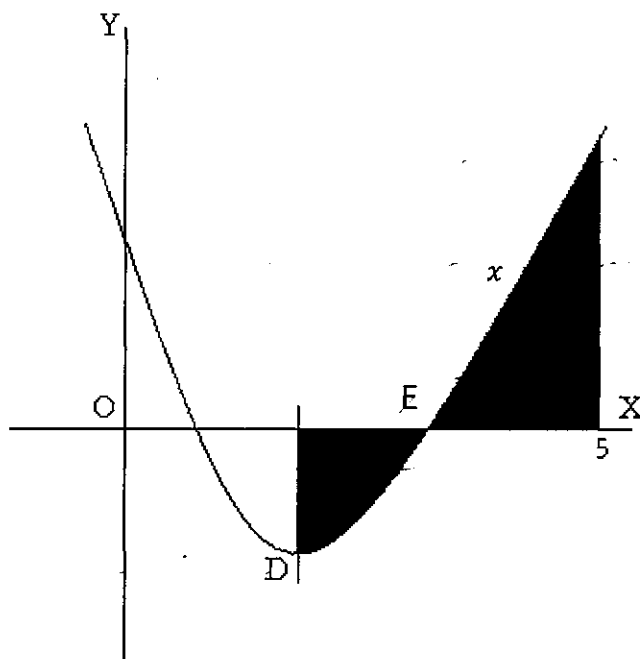
- a) Evaluate: $\lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{x - 1}$ 3
- b) Differentiate by first principle: $f(x) = x^2 + 3x$ 5
- c) Answer the following:
- 1) Find $\frac{dy}{dx}$; if $y = \frac{2x^2 - 3x + 5}{x}$ 4
- 2) Find $f'(x)$ if $f(x) = \sqrt{x}(x^2 - 3x + 4)$ 4
- d) Determine the equation of the tangent to the curve of $g(x) = \sqrt{x} + 3$ at $x = 4$. 6
- e) If $g(x)$ has the following the properties, draw a neat sketch graph of $g(x)$
 $g(0) = 32$, $g(4) = 0$, $g(-2) = 0$, $g'(0) = 0$, $g'(4) = 0$
 $g'(x) > 0$ if $x < 0$ or $x > 4$
 $g'(x) < 0$ if $0 < x < 4$ 5

QUESTION 8**15 MARKS**

- a) Find the following indefinite integrals:
1. $\int (x^3 + 5x^2 + 4) dx$ 4
2. $\int (\frac{1}{x^2} - x^3) dx$ 5
- b) Evaluate, showing all steps in the mathematical process (NO calculators to be used): 6
- $\int_4^5 x^2(3 - x) dx$

QUESTION 9

14 MARKS

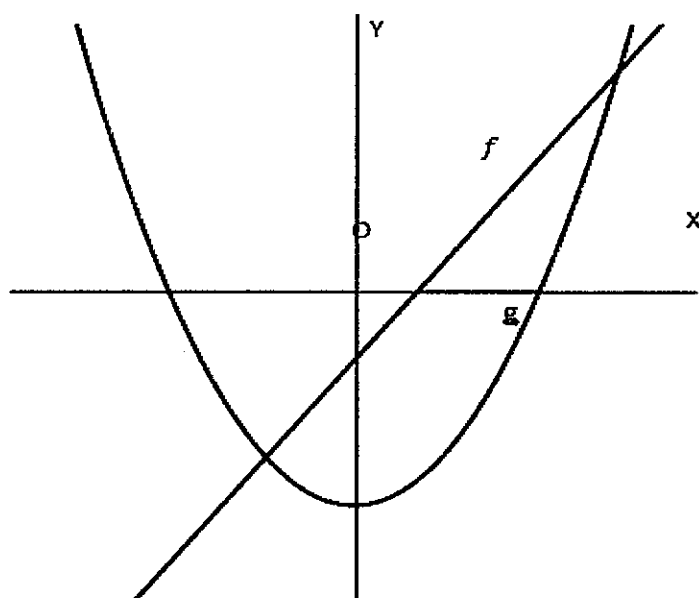


Given: $f(x) = 2x^2 - 8x + 6$

- | | |
|---|---|
| a) Determine the coordinates of D, the turning point of the parabola. | 3 |
| b) Determine the coordinates of E the x – intercept. | 3 |
| c) Use any method to determine the shaded area between D and $x = 5$ | 8 |

QUESTION 10

8 MARKS



If $f(x) = 2x - 1$ and $g(x) = x^2 - 4$, determine the size of the shaded area.

8

Question 1

a) $x^3 = 2x(x+4)$

$$x^3 = 2x^2 + 8x$$

$$x^3 - 2x^2 - 8x = 0$$

$$x(x^2 - 2x - 8) = 0$$

$$x(x + 2)(x - 4) = 0$$

$$\underline{x=0} \text{ or } \underline{x=-2} \text{ or } \underline{x=4}$$

6

b) $|2x^2 - x| - 1 + \frac{15}{|2x^2 - x| + 2} = 5$

Let $k = |2x^2 - x|$

$$k - 1 + \frac{15}{k+2} = 5$$

$$(k-1)(k+2) + 15 = 5(k+2)$$

$$k^2 + k - 2 + 15 = 5k + 10$$

$$k^2 - 4k + 3 = 0$$

$$(k - 1)(k - 3) = 0$$

$$k = 1 \text{ or } k = 3$$

$$|2x^2 - x| = 1$$

$$2x^2 - x - 1 = 0$$

$$\text{or } 2x^2 - x + 1 = 0$$

$$(2x+1)(x-1) = 0$$

$$(2x+1)(x-1) = 0$$

$$(2x-1)(x+1) = 0$$

$$(2x-3)(x+1) = 0$$

$$\underline{x = -\frac{1}{2}} \text{ or } \underline{x = 1}$$

No soln

$$\underline{x = \frac{3}{2}} \text{ or } \underline{x = -1}$$

No soln

8